

Foundation (Jalapeno)

Q1. Simplify $2^3 \cdot 2^4$

- (A) 2^{3+4}
(B) 2^{3-4}
(C) $2^{3 \cdot 4}$
(D) $2^{3/4}$
(E) 2^{4-3}

(B) $1000 \cdot 2^2$

$1000+$

(C) $1000 \cdot 2^{2-1}$

$1000 \cdot 2^2$

(D)

$1000 \cdot 2^{2+1}$

(E)

$1000 \cdot 2^{2-2}$

Q2. If $x^2 = 16$, then x^4 is

- (A) 16^2
(B) $16^{1/2}$
(C) 4^2
(D) 2^4
(E) 8^2

Q6. Simplify $4^3 \cdot 4^2$

- (A) 4^{3+2}
(B) 4^{3-2}
(C) $4^{3 \cdot 2}$
(D) $4^{3/2}$
(E) 4^{2-3}

Q3. The stock price of XYZ Inc. increases by a factor of 2^2 each year. If it was

10

last year, what will it be in two years?

(A)

$10 \cdot 2^2$

(B)

$10 \cdot 2^4$

(C)

$10 \cdot 2^{2 \cdot 2}$

(D)

$10 \cdot 2^{2+2}$

(E)

$10 \cdot 2^{2-2}$

Q7. The tax on a stock is 2^3

(A)

$256 \cdot 2^3$

(B)

$256 \cdot 2^{3-1}$

(C)

$256 \cdot 2^{3+1}$

(D)

$256/2^3$

(E)

$256/2^{3-1}$

Q4. Simplify $(3^2)^3$

- (A) 3^{2+3}
(B) $3^{2 \cdot 3}$
(C) 3^{2-3}
(D) $3^{2/3}$
(E) 3^{3-2}

Q8. Simplify $(2^2)^4$

- (A) 2^{2+4}
(B) $2^{2 \cdot 4}$
(C) 2^{2-4}
(D) $2^{2/4}$
(E) 2^{4-2}

Q5. A savings account earns interest at a rate of 2^2

(A)

$1000+$

Open Ended Questions (FRQ)

Q9. A company's profit increases by a factor of 3^2 each quarter. If the profit was

2500

last quarter, what will it be in three quarters? Show your work and simplify your answer.

Q10. Simplify the expression $2^3 \cdot 4^2 \cdot 8^1$. Show your work and explain each step.

Chef's Tips

- Tip 1: Make sure to check the units of the answers
- Tip 2: Use the rules of exponents to simplify the expressions
- Tip 3: Use real-world examples to help students understand the concepts